

Table 3.14 (continued)

	Average Asset Weights			
	U.S. Govt Bonds	U.S. Corp Bonds	U.S. Stocks	International Stocks
Risk Parity (Variance)	0.51	0.36	0.07	0.06
Risk Parity (Volatility)	0.97	−0.30	0.17	0.16
Minimum Variance	1.41	−0.51	0.07	0.03
Equal Risk Contributions	0.50	0.42	0.25	−0.17
Kelly Rule	1.18	−0.29	0.07	0.04
Proportional to Sharpe Ratio	0.24	0.21	0.21	0.35

The simple Equal Weight strategy does very well with a Sharpe ratio of 0.54. What a contrast with this strategy versus the complex mean-variance portfolio (with a Sharpe ratio of 0.07)! The Equal Weight strategy also outperforms the market portfolio (with a Sharpe ratio of 0.41). De Miguel, Garlappi, and Uppal (2009) find that the simple $1/N$ rule outperforms a large number of other implementations of mean-variance portfolios, including portfolios constructed using robust Bayesian estimators, portfolio constraints, and optimal combinations of portfolios which I covered in section 4.2. The $1/N$ portfolio also produces a higher Sharpe ratio than each individual asset class position. (U.S. bonds had the highest Sharpe ratio of 0.47 in the sample.)

Risk Parity does even better than $1/N$. The outperformance, however, of the plain-vanilla Risk Parity (Variance) versus Equal Weights is small. Risk Parity (Variance) has a Sharpe ratio of 0.59 compared to the 0.54 Sharpe ratio for Equal Weights. Risk Parity based on volatility does even better and has the highest out-of-sample Sharpe ratio of all the strategies considered, at 0.65. When risk parity strategies are implemented on more asset classes (or factor strategies, see chapter 7) in practice, historical Sharpe ratios for risk parity strategies have often exceeded one.

The outperformance of the Minimum Variance portfolio versus standard mean-variance weights and the market portfolio has been known for at least twenty years.³² One reason that minimum variance portfolios outperform the market is that there is a tendency for low volatility assets to have higher returns

³² At least since Haugen and Baker (1991).